

Algebra II with Trigonometry

Chapter 6.5

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Certainly! Here are **10 mental math questions** testing core concepts from the PDF on inverse trigonometric functions. Each can be answered without pencil and paper, and targets deep understanding or key facts.

1. Inverse Sine Range

Q: What is the range of the inverse sine function, $\sin^{-1}(x)$?

2. Cancelling with Sine

Q: If $\sin^{-1}(\sin(x)) = x$, what values of x is this true for?

3. Simple Evaluation #1

Q: What is $\sin^{-1}(1)$?

4. Inverse Cosine Domain

Q: What is the domain of the inverse cosine function, $\cos^{-1}(x)$?

5. Cancelling with Cosine

Q: If $\cos^{-1}(\cos(x)) = x$, what range of x is this true for?

6. Simple Evaluation #2

Q: What is $\tan^{-1}(0)$?

7. Range of Inverse Tangent

Q: What is the range of the inverse tangent function, $\tan^{-1}(x)$?

8. Simple Evaluation #3

Q: What is $\cos^{-1}(1)$?

9. Cancelling Identity

Q: Does $\tan(\tan^{-1}(x)) = x$ for all real x ?

10. Arc Sine Value

Q: What is $\sin(\sin^{-1}(-1/2))$?

Let me know if you'd like answers or explanations for these!